ERFAN HOSSEINI SERESHGI

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PROFESSIONAL EXPERIENCE

Postdoctoral Fellow, The Murphy Institute, Tulane University — **Aug 2025 – present**
New Orleans, LA

Graduate Research Assistant, Tulane University — **Sep 2021 – May 2025**
New Orleans, LA

Data Scientist, Tulane University (LA-CEAL NIH Project) — **Jan 2021 – Nov 2021**
New Orleans, LA

Research Software Engineer, Tulane School of Public Health — **Jan 2021 – Nov 2021**
New Orleans, LA

IT Specialist, Tulane Pre-college Programs — **Summer 2020**
New Orleans, LA

Jr. Product Manager, AIESEC — **Aug 2017 – Feb 2018**
Tehran, IR

Front-end Developer Intern, Moduland.ir — **Summer 2017**
Tehran, IR

EDUCATION

Tulane University – Ph.D. in Computer Science — **2018 – 2025**
New Orleans, LA

Amirkabir University of Technology – B.Sc. in Computer Science — **2014 – 2018**
Tehran, IR

RESEARCH AND PUBLICATIONS

Drawing Reeb Graphs — **2025**
(E. Chambers, B.T. Fasy, E. Hosseini Sereshgi, M. Löffler),
36th International Workshop on Combinatorial Algorithms (IWOCA 2025)

MAGIC: Map And Geographic Information Construction, Comparison and Conflation — **2025**
(E. Hosseini Sereshgi),
Ph.D. Dissertation, Tulane University

ROADSTER: Improved Algorithms for Subtrajectory Clustering and Map Construction — **2025**
(K. Buchin, M. Buchin, J. Gudmundsson, J. Hendriks, E. Hosseini Sereshgi, R. I. Silveira, J. Sleijster, F. Staals, C. Wenk), Computers and Geosciences Journal, Vol. 196, 49 pages.

Length-preserving Matching for Closed Curves — **2024**
(E. Hosseini Sereshgi, M. Löffler, F. Staals, C. Wenk), 31st Fall Workshop on Computational Geometry, 5 pages.

Map Stitcher: Graph Sampling-based Map Conflation **2024**
(E. Hosseini Sereshgi, and C. Wenk), 3rd ACM SIGSPATIAL International Workshop on Spatial Big Data and AI for Industrial Applications (GeoIndustry), 11 pages.

Graph Sampling for Map Comparison **2024**
(J. Aguilar, K. Buchin, M. Buchin, E. Hosseini Sereshgi, R.I. Silveira, C. Wenk), ACM Transactions on Spatial Algorithms and Systems 10(3), 24 pages.

Drawing Reeb Graphs **2023**
(E. Chambers, B.T. Fasy, E. Hosseini Sereshgi, M. Löffler, S. Percival), 31st International Symposium on Graph Drawing and Network Visualization, Poster paper.

On Length-sensitive Fréchet Similarity **2023**
(K. Buchin, B.T. Fasy, E. Hosseini Sereshgi, C. Wenk), 18th Algorithms and Data Structures Symposium (WADS): 208-231.

Merging Roadmaps using Graph Distance Measures **2022**
(E. Hosseini Sereshgi and C.Wenk), 30th Fall Workshop on Computational Geometry, 5 pages.

Graph Sampling for Map Comparison (received best paper award) **2021**
(J. Aguilar, K. Buchin, M. Buchin, E. Hosseini Sereshgi, R.I. Silveira, C. Wenk), 3rd ACM SIGSPATIAL International Workshop on Spatial Gems.

Measuring Length-Preserving Fréchet Correspondence for Graphs in R^2 **2021**
(K. Buchin, B. T. Fasy, E. Hosseini Sereshgi and C. Wenk), 29th Fall Workshop on Computational Geometry, 5 pages.

Improved Map Construction using Subtrajectory Clustering **2020**
(K. Buchin, M. Buchin, J. Gudmundsson, J. Hendriks, E. Hosseini Sereshgi, V. Sacristán, R. Silveira, J. Sleijster, F. Staals, C. Wenk), 4th ACM SIGSPATIAL Workshop on Location-Based Recommendations, Geosocial Networks, and Geoadvertising: article 5, 4 pages.

Computing Relevant Subtrajectory Bundles Faster **2020**
(E. Hosseini Sereshgi and C. Wenk), SoCG, Computational Geometry: Young Researchers Forum, 2 pages.

Clustering Gene Expression with Polygonal Chain Alignment **2018**
(H. Banaee, E. Hosseini Sereshgi, A. Mohades, F. Zare)
B.Sc. capstone project, Amirkabir University of Technology

PRESENTATIONS

Length-preserving Matching for Closed Curves **2024**
31st Fall Workshop on Computational Geometry (FWCG)

Map Stitcher: Graph Sampling-based Map Conflation **2024**
3rd ACM SIGSPATIAL International Workshop on Spatial Big Data and AI for Industrial Applications (GeoIndustry)

On Length-sensitive Fréchet Similarity **2023**
18th *Algorithms and Data Structures Symposium (WADS)*

Merging Roadmaps using Graph Distance Measures **2022**
30th Fall Workshop on Computational Geometry (FWCG)

Graph Sampling for Map Comparison **2021**
3rd ACM SIGSPATIAL International Workshop on Spatial Gems

Measuring Length-Preserving Fréchet Correspondence for Graphs in $\mathbb{R}^2$ **2021**
29th Fall Workshop on Computational Geometry (FWCG)

The Study of Gentrification on Social Urban Simulation - How Income and Interest Can Shape Neighborhoods **2020**
Tulane University

Improved Map Construction using Subtrajectory Clustering **2020**
4th ACM SIGSPATIAL, Location-based Recommendations, Geosocial Networks and Geoadvertising

Computing Relevant Subtrajectory Bundles Faster **2020**
36th International Symposium on Computational Geometry (SoCG), Young Researchers Forum

Clustering Gene Expression with Polygonal Chain Alignment **2018**
Amirkabir University of Technology

A brief Intro to Computational Geometry **2017**
Amirkabir University of Technology, Graduate studies seminar

TEACHING EXPERIENCE

Algorithms (Graduate level) **Fall 2024**
Teaching assistant

Computational Geometry (Graduate level) **Fall 2023**
Teaching assistant

Arduino course at Tulane Pre-college Program **Summer 2022**
Instructor

Electrical Engineering course at Tulane Pre-college Program **Summer 2022**
Instructor

Introduction to Discrete Math **Fall 2020**
Teaching assistant

Introduction to Algorithms **Fall 2019**
Teaching assistant

Python game design at Tulane Pre-college Program **Summer 2019 and 2022**
Instructor

Robotics at Tulane Pre-college Program **Summer 2019 and 2022**
Instructor

Intro to Computer Science 1 (Python) **Fall 2018, Spring 2019, Spring 2020**
Teaching assistant

Operating systems Lab/Workshop **Spring 2017**
Teaching assistant

C++ Programming teacher at Helli 4 high school **2014 – 2015**
Instructor

LEADERSHIP & VOLUNTEER EXPERIENCE

- Tulane Computer Science Graduate Student Council (CSGSC) founding member
- Tulane Computer Science Graduate Student Council department representative (Spring 2023)
- Tulane Graduate Studies Student Association Travel fund committee member (2023)
- Senator at Tulane Graduate and Professional Student Association (2020-2022)
- Tulane Graduate Studies Student Association representative (2020-2021)
- IT team leader at AIESEC in University of Tehran (2017-2018)
- Marketing designer at AIESEC in Amirkabir University of Technology (Spring 2017)
- AIESEC global volunteer for raising public awareness about clean energy and recycling in Guangzhou, China. (Summer 2016)
- Member of scientific association of math and computer science at Amirkabir University of Technology (2015-2016)

CERTIFICATES

- Fundamentals of Deep Learning from Nvidia
- Java programming from Amirkabir University of Technology
- Android development from Amirkabir University of Technology
- Web development and web design from Amirkabir University of Technology
- CITI Group1. Biomedical Researchers
- CITI Group4. IRB, Biomedical Research

SKILLS & ABILITIES

- Skilled in Python, C++ and Java
- Familiar with HTML, CSS, Javascript and SQL
- Have worked with R and R studio
- Familiar with Git and Visual Paradigm
- Have some intermediate knowledge about Android Studio
- Have experience working with Adobe Photoshop and illustrator
- Familiar with QGIS, ArcGIS and GDAL libraries
- Familiar with Pytorch, Keras and Scikit-Learn

LANGUAGES

- English (fluent)
- Persian (native)

- Arabic (intermediate)

HONORS & AWARDS

- Tulane Connolly Alexander Institute for Data Science Summer Graduate Award (2023)
- Best paper award at ACM SIGSPATIAL: Spatial Gems (2021)
- Ranked 6th in the Iranian national CS graduate school entrance exam (2018)
- Ranked among top 5 computer science students at Amirkabir University of Technology (class of 2018)
- Semi-finalist in 2014 BAYAN coding contest in Iran
- Ranked among 3% in the Iranian national university/college entrance exam (2014) (More than 60,000 students)
- Was selected by and studied at the national organization for development of exceptional talents (NODET) in Iran